

Executive Data Cleaning Report

Weekly Project: Titanic Dataset Preparation & Quality Assurance

Dataset: Titanic-Dataset.csv Output File: cleaned_titanic.csv Status: Ready for Analysis

891

TOTAL ROWS

866

MISSING VALUES
FIXED

0

DUPLICATES FOUND

100%

DATA
COMPLETENESS

1. DATA QUALITY ISSUES IDENTIFIED

- **Extensive Missingness in Cabin Data:** The Cabin column had 687 missing values (~77.1% missing), making direct analytical modeling without preprocessing impossible.
- **Moderate Null Values in Passenger Age:** The Age feature contained 177 missing entries (~19.9%), requiring imputation before any age-based demographics analysis.
- **Minor Null Values in Port of Embarkation:** The Embarked feature had 2 missing records (~0.2%).
- **Unoptimized Column Naming & Data Types:** Headers used inconsistent uppercase naming, and categorical fields (Survived, Pclass, Sex, Embarked) were stored as unoptimized integer/string formats.

2. CLEANING TECHNIQUES APPLIED

- **Median Imputation for Age:** Replaced all 177 missing Age entries with the global median age (28.0 years) to maintain central tendency without introducing extreme outliers.
- **Mode Imputation for Embarked:** Imputed the 2 missing Embarked records with ' S ' (Southampton), which represents the statistical mode of the feature.
- **Explicit Categorical Placeholder:** Replaced null values in Cabin with ' Unknown ' to preserve passenger row integrity without dropping 77% of dataset records.
- **Standardization & Type Conversion:** Standardized column titles to lowercase snake_case and cast categorical variables to explicit Pandas category dtypes for memory optimization.

3. ASSUMPTIONS MADE DURING PREPROCESSING

- **Missingness Mechanism:** Assumed age data was missing at random (MAR), making median imputation a statistically safe technique for preserve distribution.
- **Cabin Feature Retention:** Assumed missing cabin numbers indicate standard unrecorded accommodation rather than corrupted data, preserving rows for general survival modeling.

4. KEY OBSERVATIONS FROM THE DATASET

- **Dataset Integrity:** The dataset contains 891 individual passenger records with zero duplicate rows detected.
- **Data Readiness:** Post-cleaning verification confirms 100% data completeness with zero remaining null values across all 12 variables.

Summary: The preprocessed dataset `cleaned_titanic.csv` has been verified and is structured for exploratory data analysis (EDA) and machine learning workflows.